

Step-Up Fabrication Document

Top Fabrication (Scale 1:1)

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR YELLOW.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.

B. Tg 170 C OR EQUIVALENT.

C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY NA.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE85,000 × 100,000 mm

BOARD THICKNESS1,602 mm

TRACE WIDTH0,300 mm

TRACE TO TRACE-0,000 mm

MIN. HOLE (PTH)0,250 mm

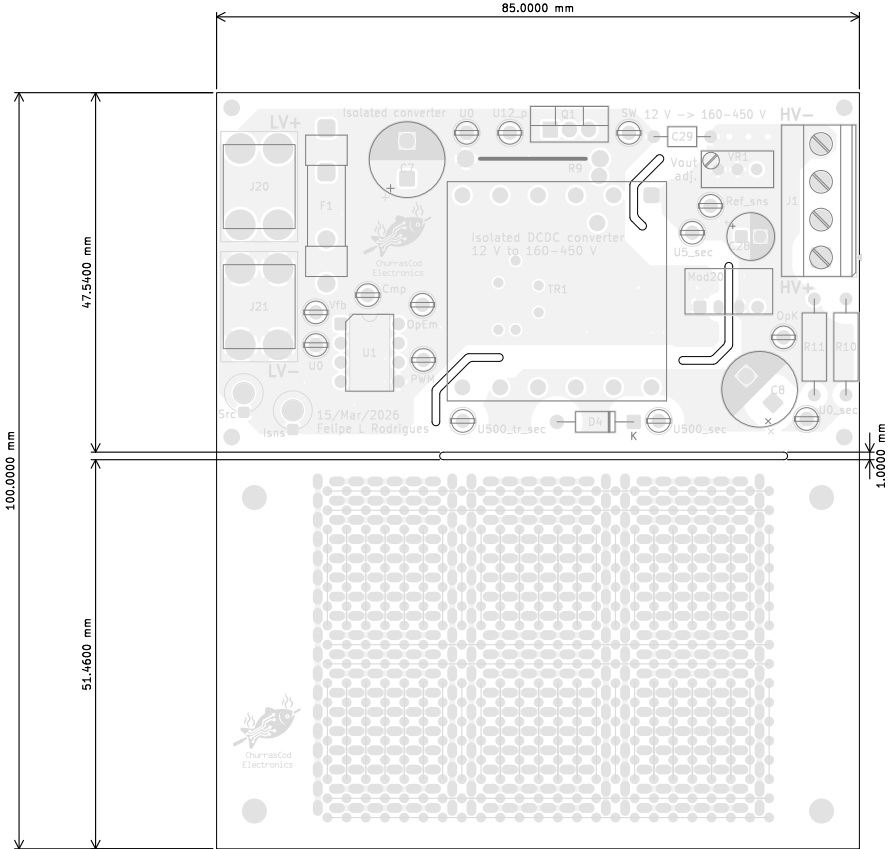
MIN. HOLE (NPTH)2,100 mm

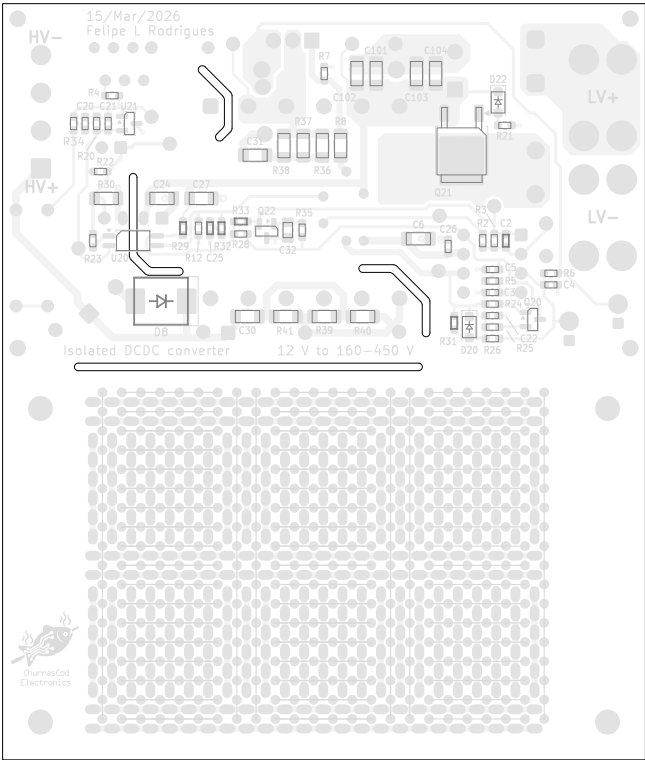
ANNULAR RING0,150 mm

COPPER TO HOLE0,200 mm

COPPER TO EDGE0,250 mm

HOLE TO HOLE0,254 mm





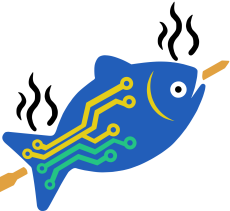
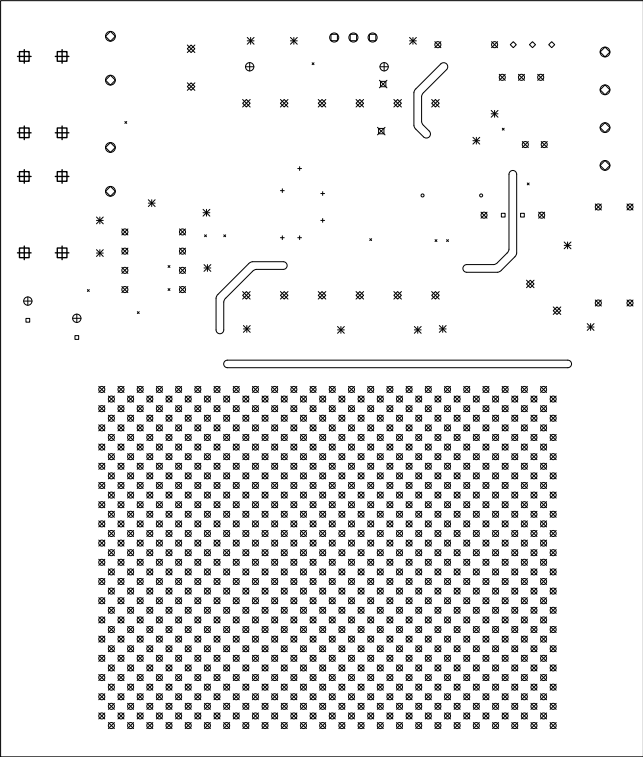
Comments:	Company: NA		Variant: CHECKED	Git Hash: 9fc4300
	Board Name: Step-Up		Project Name: Isolated step-up 12:450 V	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)
Sheet Path:		Reviewer: FR	Size: A4	Sheet: 2 of 7

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Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	13	0.25mm (9,84mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
○	2	0.40mm (15,75mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
+	6	0.50mm (19,69mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
□	4	0.70mm (27,56mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
◇	3	0.80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
⊗	885	0.80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
※	16	0.90mm (35,43mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊗	2	1.00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
⊗	16	1.00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊕	4	1.10mm (43,31mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊕	3	1.20mm (47,24mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
○	8	1.30mm (51,18mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊕	8	1.80mm (70,87mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
Total 970						



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Board Name:

Step-Up

Project Name:

Isolated step-up 12:450 V

Sheet Title:

Drill Drawing (L1 - L4)

File Name:

Isolated_12to450V.kicad_pcb

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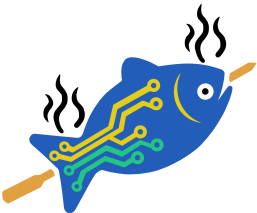
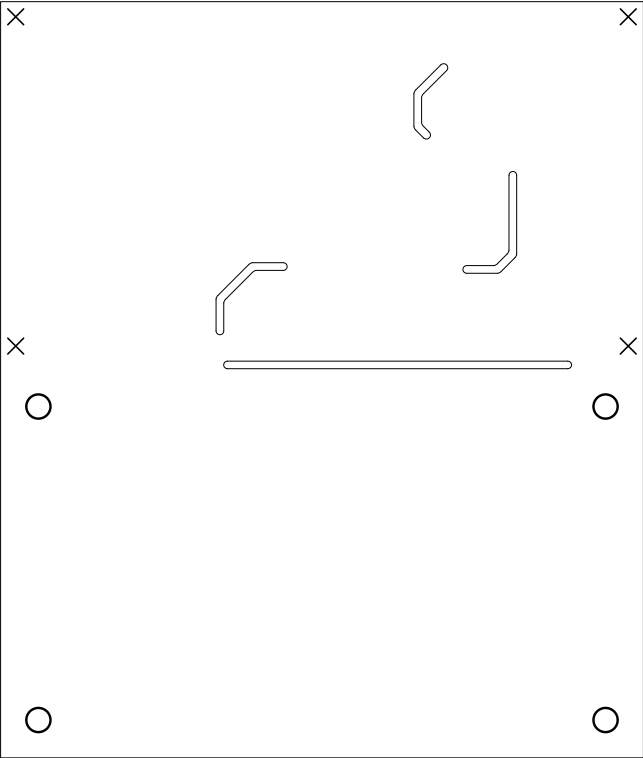
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Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	4	2,10mm (82,68mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
○	4	3,20mm (125,98mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
Total 8						



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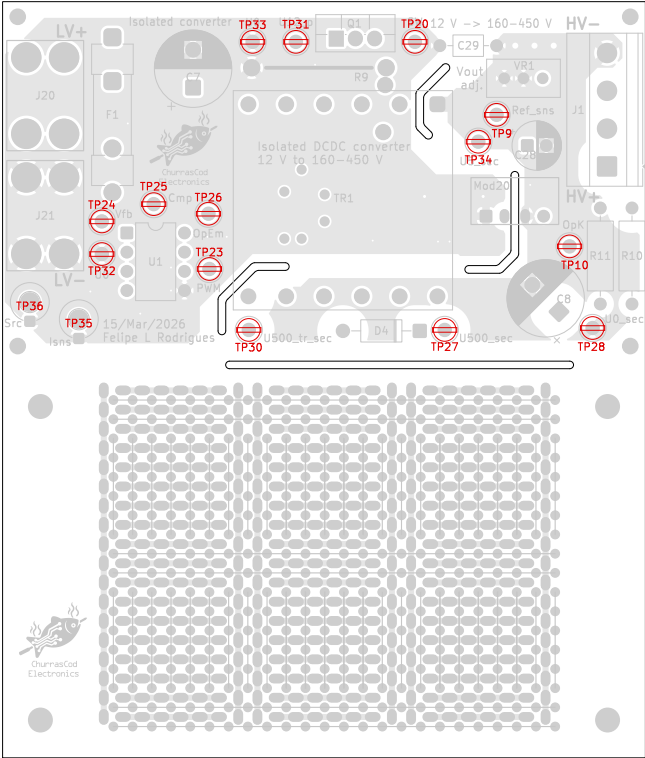
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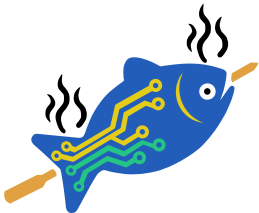
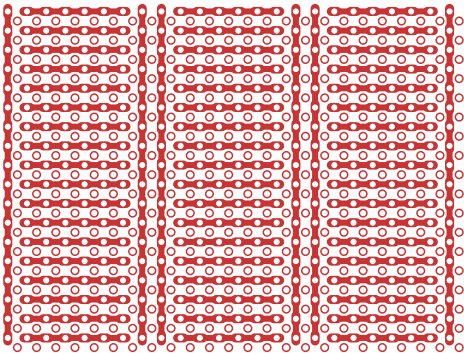
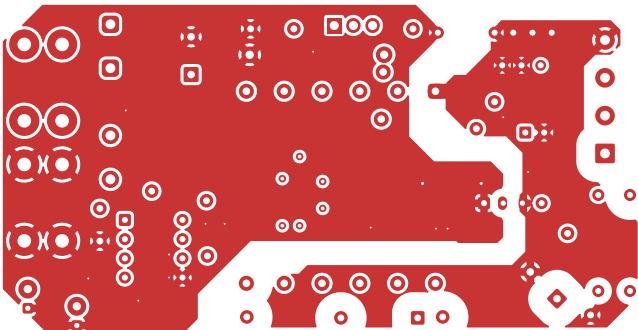
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Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP9	Sec_ref_sns	67.32	35.01
TP10	Sec_OpK	76.97	17.61
TP20	SW	56.53	44.66
TP23	PWM	29.35	14.62
TP24	Vfb	15.13	20.91
TP25	Comp	21.98	23.20
TP26	OpEm	29.22	21.93
TP27	U500_sec	60.46	6.56
TP28	U0_sec	80.02	6.81
TP30	U500_tr_sec	34.56	6.56
TP31	U12_p	40.78	44.66
TP32	U0	15.13	16.59
TP33	U0	35.06	44.66
TP34	U5_sec	64.91	31.45
TP35	Isns	12.08	6.70
TP36	Src.U0	5.60	8.97
TP102	Net-(TP101-Pad1)	20.46	-1.44
TP103	Net-(TP101-Pad1)	23.00	-1.44
TP104	Net-(TP101-Pad1)	25.54	-1.44
TP105	Net-(TP101-Pad1)	28.08	-1.44
TP106	Net-(TP101-Pad1)	30.62	-1.44
TP107	Net-(TP107-Pad1)	17.92	-3.98
TP108	Net-(TP107-Pad1)	20.46	-3.98
TP109	Net-(TP107-Pad1)	23.00	-3.98
TP110	Net-(TP107-Pad1)	25.54	-3.98
TP111	Net-(TP107-Pad1)	28.08	-3.98
TP112	Net-(TP107-Pad1)	30.62	-3.98
TP113	Net-(TP113-Pad1)	17.92	-6.52
TP114	Net-(TP113-Pad1)	20.46	-6.52
TP115	Net-(TP113-Pad1)	23.00	-6.52
TP116	Net-(TP113-Pad1)	25.54	-6.52

All dimensions are in millimeters
unless otherwise specified.



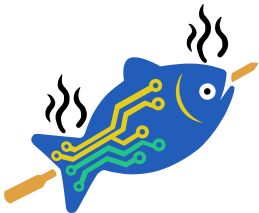
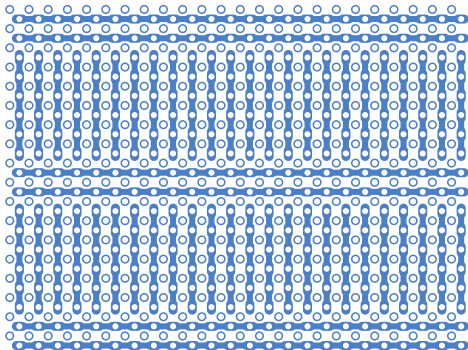
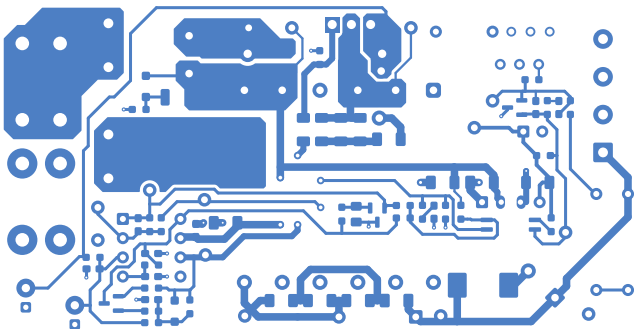


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B.Cu (Scale 1:1)



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	File Name:	Designer:	Date:	Revision:
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Sheet Path:		Reviewer:	Size:	Sheet:
		FR	A4	7 of 7